

KALAN FERGUSON

kalanferguson.com

kalanferguson005@gmail.com | 0426 060 934 | Brisbane, QLD | [linkedin.com/in/kalan-ferguson-288858326](https://www.linkedin.com/in/kalan-ferguson-288858326)

Australian Citizen | NV1 Security Clearance Eligible

CAREER OBJECTIVE

Third-year Mechanical Engineering (Honours) / Aerospace Minor student at QUT (Major GPA 6.4/7.0). Experienced in structural analysis, FEA, and independent mechanical design (F-35B vectoring nozzle). Seeking 2026 Mechanical Engineering internship.

EDUCATION

Bachelor of Mechanical Engineering (Honours) | Aerospace Minor | QUT Expected Dec 2027

- Year 2 Major GPA: 6.4/7.0 | Cumulative GPA: 5.8/7.0
- QUT Executive Dean's Commendation for Academic Excellence (awarded for 6.5+ semester GPA)
- High Distinctions: Thermodynamics, Foundations of Electrical Engineering, Mechanical Design, Engineering Mathematics & Statistics
- Current units: Aircraft Systems & Flight, Dynamics of Machines, Design of Machine Elements, Fluid Mechanics

TECHNICAL PROJECTS

F-35B 3-Bearing Swivel Duct / *Independent Mechanical Study* 2025

- Independently designed and motion-validated a fully parametric SolidWorks model of the F-35B 3-Bearing Swivel Duct, capturing non-orthogonal rotational synchronisation across the full thrust-deflection range.
- Simulated vectoring motion through SolidWorks Motion Study, verifying geometric clearance and range-of-motion at each deflection stage.
- Resolved mechanical interference and volumetric constraints within the swivel junction through iterative geometric refinement of the bearing interfaces.

Weld Fatigue Design Audit / *EGB316 - Design of Machine Elements* 2026

- Conducted full fatigue audit of a critical welded suspension bracket using first-principles analysis and ANSYS FEA, achieving infinite fatigue life (FoS = 6.33).
- Characterised 100x50 mm fillet weld group from first principles; applied Modified Goodman analysis with $K_f = 2.7$ and Marin-modified C300 endurance limit; yield FoS = 14.46 under normal operation.
- ANSYS mesh convergence from 1 mm to 0.1 mm at weld toe; frictionless boundary condition modelling reduced peak stress 40% vs fixed constraint, quantifying conservatism in the analytic solution.

Fixed Wing Autonomous UAV / *QUT Aerospace Society - Aerostructures Lead* 2025 - Present

- Aerostructures Lead on QUT Aerospace Society team developing a fully autonomous fixed-wing UAV from blank-sheet; leading configuration trade studies and airfoil selection ($Re \sim 110,000$).
- Led trade studies evaluating boom-and-pod, flying wing, and conventional layouts against stability, structural efficiency, and manufacturing complexity criteria.
- Recommended NACA 4412 following evaluation of Selig, Eppler, Drela, and NACA families at $Re \sim 110,000$; selected for Cl/Cd performance, gentle stall characteristics, and spar integration depth.

3-Stage Planetary Gearbox / *Design, Analysis & Physical Testing* 2025

- Designed and physically tested a 3-stage planetary gearbox to 20 Nm; ANSYS FEA and analytical fatigue analysis across all stages; root cause attributed to housing compliance following documented sequential failure.

PROFESSIONAL EXPERIENCE

Installer Assistant | Brisbane Glass Splashbacks Sep 2024 - Aug 2025

- Maintained +/-1 mm installation tolerances across 300+ custom glass units in residential and commercial environments.
- Applied technical drawings and spatial reasoning to resolve on-site alignment and fit-up constraints; managed material integrity in high-value, risk-sensitive installations.

Warehouse Logistics | LSKD Loganholme Nov 2024 - Jan 2025

- Processed 5,000+ orders under KPI-driven throughput targets, demonstrating precision and operational reliability.

TECHNICAL SKILLS

FEA & Simulation: ANSYS Mechanical, SolidWorks Simulation

CAD: SolidWorks (parametric modelling, assemblies, motion studies)

Analysis Methods: Modified Goodman fatigue, Von Mises, Lewis factors, kinematic analysis, load path analysis

Programming: MATLAB, Python

Standards: AS 4024 weld quality grades; AS/NZS structural steel

Documentation: Technical report writing, formal design reports, engineering drawings

ADDITIONAL ACTIVITIES

- QUT Aerospace Society - Aerostructural lead on Fixed Wing Autonomous UAV project; attending industry events and technical seminars.
- Independent technical research in Australian sovereign defence platforms; published analysis of MQ-28A Ghost Bat programme and AUKUS Pillar II implications.